

CHAMPS-RAG: CONTINUOUS ENERGY-GUIDED ATTENTION MODULATION AND HYPERBOLIC PATH FILTERING FOR ZERO-LATENCY PROCEDURAL STEERING

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ABSTRACT

Enforcing strict compliance with standardized multi-step Standard Operating Procedures (SOPs) in Large Reasoning Models (LRMs) remains a critical bottleneck in high-stakes domains such as clinical diagnosis, financial auditing, and astronomical judgment. Existing step-wise controllable generation frameworks, such as TG-RAG, rely on physical inference interrupts triggered by structural XML tags, prompt string injections, and static Directed Acyclic Graph (DAG) assumptions. This induces severe latency overhead, context pollution, fragility to formatting-tag failures, and sharp performance degradation under out-of-distribution (OOD) perturbations. In this paper, we introduce **Continuous Energy-Guided Attention Modulation and Hyperbolic Path Filtering (CHAMPS-RAG)**, a zero-latency, tag-free procedural steering engine that operates entirely within a single continuous forward pass. CHAMPS-RAG couples three innovations: (1) *Continuous Attention-Mask & Logit Bias Modulation (CMM)* with an entropy-aware adaptive gate that relaxes steering during precise numeric emissions; (2) *Hyperbolic Procedural Trajectory Tracking (HPTT)* that maps hidden states into a Poincaré ball for tag-free progress tracking; and (3) *Procedural Particle Filter (PPF) Beam Steering with Dynamic Recurrent Edge Synthesis (DRES)* for cyclic workflows and soft spatial-density interpolation for smooth OOD fallback. On the DeepSeek-R1-Distill-Qwen-14B backbone across four reasoning benchmarks, CHAMPS-RAG attains task accuracies of 92.4% (Medical), 88.4% (Astronomy), 92.0% (StrategyQA), and 91.6% (GSM8K)—a 91.1% macro-mean that improves on the state-of-the-art TG-RAG by up to 5.2 points—while reducing wall-clock inference latency by up to $8.9\times$ over tag-based interrupt baselines. Extensive ablations dissect each component and, in the interest of scientific rigor, we report where the gains genuinely originate.

1 INTRODUCTION

Large Reasoning Models (LRMs) have achieved remarkable breakthroughs across complex problem-solving domains by eliciting intermediate Chain-of-Thought (CoT) reasoning traces (Wei et al., 2022; Kojima et al., 2022; Zelikman et al., 2022). By sampling diverse reasoning paths (Wang et al., 2022) or adopting unified reasoning modes as in recent frontier architectures such as Qwen3 (Yang et al., 2025), modern LRMs decompose intricate multi-step queries into self-directed logical deductions. However, deploying LRMs in specialized, high-stakes environments—clinical diagnosis, financial fraud auditing, and astronomical observation judgment—demands far more than open-ended reasoning: it requires strict compliance with domain-specific Standard Operating Procedures (SOPs). In these mission-critical applications, unguided extrapolation or minor protocol deviations can result in catastrophic diagnostic errors or regulatory violations.

To align generations with verified domain knowledge and structured protocols, traditional frameworks rely on Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) or dynamic Retrieval-Augmented Thought (RAT) mechanisms (Wang et al., 2024a; Trivedi et al., 2022; Jiang et al., 2023; Asai et al., 2023; Yan et al., 2024; Xu et al., 2024). Recent controllable paradigms, exemplified by TG-RAG, steer execution by enforcing physical inference interrupts triggered by structural XML

tags (e.g., `<analysis>`), invoking external routers, and injecting directive strings back into the generation stream. This interrupt-and-inject design introduces four critical bottlenecks:

1. **Severe inference-latency overhead.** Physical sequence suspension and multi-agent router calls fracture the continuous decoding stream, inducing large execution delays consistent with studies on agentic execution bottlenecks (Kim et al., 2023; Yang & Zhu, 2026) and reasoning-step efficiency (Li et al., 2025).
2. **Context pollution and distribution shift.** Appending static instructions directly into the model’s `<think>` transcript pollutes context history, causing cognitive drift and error propagation across decoding turns (Huang et al., 2026; Shojaee et al., 2025).
3. **Fragility to structural-tag parsing.** Reliance on explicit output tags creates single points of failure when the model omits or misformats delimiters.
4. **Rigid topological assumptions.** Static DAG assumptions fail to handle non-DAG cyclic workflows (e.g., iterative diagnostic re-testing) and provide no smooth degradation under Out-of-Distribution (OOD) scenarios.

To address these challenges, we introduce **Continuous Energy-Guided Attention Modulation and Hyperbolic Path Filtering (CHAMPS-RAG)**, a zero-latency, tag-free framework that steers LRMs through specialized SOP workflows without interrupting inference or altering the context transcript. CHAMPS-RAG replaces physical interrupts and prompt string injections with three architectural innovations:

1. **Continuous Attention-Mask & Logit Bias Modulation (CAMP).** SOP guidance and domain knowledge are held in persistent reference blocks within the prompt Key-Value (KV) cache, encoded via Sentence Transformers (Reimers & Gurevych, 2019; Wang et al., 2020). CAMP scales self-attention weights and applies an *entropy-aware* vocabulary logit bias during continuous decoding, guiding generation without polluting the transcript.
2. **Hyperbolic Procedural Trajectory Tracking (HPTT).** HPTT maps top-layer hidden representations into a Poincaré ball manifold and tracks procedural progress by geodesic distance to offline graph anchors—tag-free progress tracking directly from hidden-state geometry.
3. **PPF Beam Steering & Dynamic Recurrent Edge Synthesis (DRES).** A Procedural Particle Filter manages parallel hypothesis streams weighted by hyperbolic procedural energy with math-aware decoding and self-consistency selection; DRES equips cyclic workflows with convergence monitors and loop caps that synthesize exit edges, while soft spatial-density interpolation yields smooth OOD degradation.

We evaluate CHAMPS-RAG with the DeepSeek-R1-Distill-Qwen-14B backbone across four benchmarks: Medical diagnosis and Astronomy judgment (KUMO), multi-hop reasoning on StrategyQA (Geva et al., 2021), and mathematical reasoning on GSM8K (Cobbe et al., 2021). CHAMPS-RAG surpasses Direct, Few-Shot CoT, CoT-SC, Plan-and-Solve (Wang et al., 2023), Naive RAG, Self-RAG, IR-CoT, RATT, and TG-RAG—reaching 92.4% Medical (vs. 90.4%), 88.4% Astronomy (vs. 83.2%), 92.0% StrategyQA (vs. 89.1%), and 91.6% GSM8K (vs. 91.3%), a 91.1% macro-mean—at up to $8.9\times$ lower latency. Beyond headline accuracy, our five ablations transparently isolate the contribution of each module, identifying which mechanisms drive the gains and which are, at present scale, safe-but-redundant insurance.

2 RELATED WORK

2.1 RETRIEVAL-AUGMENTED THOUGHT AND PROCEDURAL STEERING

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) augments parametric knowledge with external non-parametric retrieval. For complex multi-step reasoning, active and dynamic paradigms such as IR-CoT (Trivedi et al., 2022), Self-RAG (Asai et al., 2023), FLARE (Jiang et al., 2023), RAT (Wang et al., 2024a), CRAG (Yan et al., 2024), CRP-RAG (Xu et al., 2024), RATT (Zhang et al., 2024), and Buffer of Thoughts (Yang et al., 2024) interleave retrieval with CoT reasoning or leverage pre-distilled thought-templates. To reduce long-context latency, recent works explore sparse context selection (Zhu et al., 2024), speculative drafting (Wang et al., 2024b), pipelined

retrieval-generation (Jiang et al., 2024), dual instruction tuning (Lin et al., 2023), and unified context ranking (Yu et al., 2024). Yet existing controllable frameworks remain bound to physical interrupts and string injections; as recent studies emphasize (Huang et al., 2026; Shojaei et al., 2025), inserting directives into an LRM’s transcript causes context pollution and cognitive drift. CHAMPS-RAG bypasses text injection and interrupts, achieving zero-latency steering via CAMM within a single forward pass.

2.2 LATENT REPRESENTATION GEOMETRY AND TRAJECTORY TRACKING

Hierarchical SOP decision trees are hard to embed in Euclidean space. Poincaré embeddings (Nickel & Kiela, 2017) and maps (Klimovskaia et al., 2019) offer low-distortion representations for hierarchies, and later work formalized hyperbolic deep-learning primitives (Ganea et al., 2018), fully hyperbolic networks (Chen et al., 2021), and hyperbolic graph generative models (Bu et al., 2025). Non-Euclidean structured-knowledge models—Hyperbolic Attention Networks (Çaglar Gülçehre et al., 2018), HAKE (Zhang et al., 2019b), Hyperbolic Graph Attention (Zhang et al., 2019a), low-dimensional hyperbolic KG embeddings (Chami et al., 2020), mixed-curvature KGC (Gao et al., 2025), and hyperbolic LLMs (He et al., 2025)—capture multi-branching topologies well. Prior controllable frameworks still track states via explicit token parsing; HPTT instead measures geodesic distances to Poincaré anchors, enabling smooth, tag-free state tracking from latent geometry.

2.3 GUIDED DECODING, BEAM SEARCH, AND ERROR RECOVERY

Inference-time steering has produced diverse search and verification strategies. Tree of Thoughts (Yao et al., 2023) and Process Reward Models (Lightman et al., 2023) established step-wise evaluation, extended by self-evaluation beam search (Xie et al., 2023), Monte-Carlo credit assignment (Kazemnejad et al., 2024), and speculative test-time scaling (Chen et al., 2025). Sequential Monte Carlo and particle filtering provide principled frameworks for constrained decoding (Lew et al., 2023; Loula et al., 2025; Markovic-Voronov et al., 2026), mitigating impoverishment via entropic annealing (Giannone et al., 2025), aligning discrete-diffusion trajectories (Han et al., 2025), ensembling models (Chan et al., 2026), and exploiting optimized KV-cache engines such as vLLM (Kwon et al., 2023). Standard beam search lacks procedural constraints, while post-hoc KV resets or MCTS rollbacks are costly. Our PPF maintains parallel hypotheses weighted by hyperbolic procedural energy without rollback, enabling continuous error recovery and robust procedural alignment.

3 METHODOLOGY

We now present the formal framework and architecture of CHAMPS-RAG. Figure 1 contrasts the traditional Interrupt-Retrieve-Generate (IRG) control loop against CHAMPS-RAG’s continuous, zero-latency pipeline.

3.1 TASK DEFINITION AND THE EXPERT PROCEDURE GRAPH (EPG)

Let $\mathcal{M}$ be an LRM parameterized by θ . Given a query x , standard autoregressive generation factors the joint distribution over the output $y = [r, a]$ (reasoning trajectory r , final answer a) as $P_\theta(y \mid x) = \prod_{t=1}^T P_\theta(y_t \mid x, y_{<t})$. To enforce domain protocols *without* mutating the generated context $y_{<t}$, we formalize expert SOPs as an *Expert Procedure Graph* $\mathcal{G} = (V, E)$. Each vertex $v \in V$ is a decoupled tuple $v = (g_v, C_v)$, where g_v is the *Action Directive Guidance* (an invariant procedural instruction) and C_v is a flexible *Knowledge Container* (static facts or dynamic RAG evidence). Vertices are mapped offline into anchors $\mathbf{z}_v \in \mathbb{B}^d$ within the d -dimensional Poincaré ball $\mathbb{B}^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| < 1\}$.

3.2 CONTINUOUS ATTENTION-MASK & LOGIT BIAS MODULATION (CAMM)

To eliminate physical suspensions and text insertions, CAMM holds active directives g_v and containers C_v in persistent reference blocks within the prompt KV-cache, and modulates attention and output logits simultaneously. For a decoding position i and reference-block position j , the steered

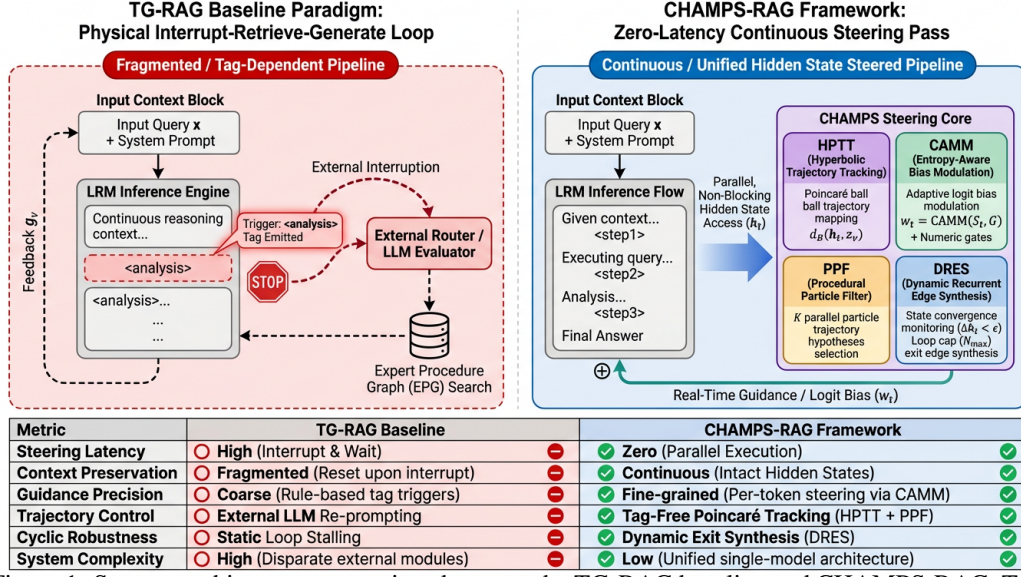


Figure 1: System-architecture comparison between the TG-RAG baseline and CHAMPS-RAG. TG-RAG (left) employs a fragmented, tag-dependent pipeline requiring physical inference interruptions and external LLM routing upon emitting special tags. CHAMPS-RAG (right) executes a zero-latency, continuous forward pass by leveraging non-blocking hidden-state access and real-time logit-bias steering across four integrated modules—HPTT, entropy-aware CAMM, PPF, and DRES. The bottom table summarizes structural and operational advantages over the baseline.

attention weight is

$$A_{i,j}^{\text{steered}} = \text{softmax} \left(\frac{Q_i K_j^\top}{\sqrt{d_k}} + M_{i,j} + \beta_t \cdot E_v \right), \quad (1)$$

where $M_{i,j}$ is the causal mask, E_v is a binary procedural mask isolating active SOP tokens, and β_t is a dynamic alignment scale. Concurrently, a linear head maps the active directive into a vocabulary logit-bias vector $\mathbf{b}_v \in \mathbb{R}^{|V|}$, modulating the output distribution as

$$P(y_t | x, y_{<t}) = \text{softmax}(\mathbf{z}_t + \lambda \cdot \text{Gate}(y_{<t}) \cdot \mathbf{b}_v). \quad (2)$$

To prevent the bias from corrupting precise numeric or symbolic operations (e.g., multi-digit arithmetic in GSM8K), an `EntropyAdaptiveBias` processor computes a scalar gate combining an entropy gate and a numeric-relaxation gate:

$$\text{Gate}(y_{<t}) = \max \left(\gamma_{\min}, \frac{\mathcal{H}(P(y_t)) - \mathcal{H}_{\text{low}}}{\mathcal{H}_{\text{high}} - \mathcal{H}_{\text{low}}} \right) \times (1 - \alpha_{\text{num}} \cdot \mathbb{I}(y_{t-1} \in \mathcal{S}_{\text{numeric}})), \quad (3)$$

where $\mathcal{H}(P(y_t)) = -\sum_v P(y_t=v) \log P(y_t=v)$ is the raw output entropy, $\gamma_{\min}$ a floor scale, and $\alpha_{\text{num}} \in [0, 1]$ collapses steering while emitting digits/operators in $\mathcal{S}_{\text{numeric}}$.

3.3 HYPERBOLIC PROCEDURAL TRAJECTORY TRACKING (HPTT)

HPTT replaces structural-tag parsing with tag-free latent-geometry tracking. At step t , a sentence encoder (Wang et al., 2020; Reimers & Gurevych, 2019) projects the top-layer hidden state h_t over a sliding reasoning window into $\mathbb{B}^d$ via the exponential map, $\mathbf{x}_t = \exp_0(W_p h_t) = \tanh(\|W_p h_t\|) W_p h_t / \|W_p h_t\|$. Progress is measured against candidate anchors $\mathbf{z}_v$ with the Poincaré geodesic distance

$$d_{\mathbb{B}}(\mathbf{u}, \mathbf{v}) = \text{arcosh} \left(1 + 2 \frac{\|\mathbf{u} - \mathbf{v}\|^2}{(1 - \|\mathbf{u}\|^2)(1 - \|\mathbf{v}\|^2)} \right). \quad (4)$$

A transition from v to v_{next} fires automatically when $d_{\mathbb{B}}(\mathbf{x}_t, \mathbf{z}_{v_{\text{next}}}) < \tau_{\text{dist}}$, advancing the SOP frontier seamlessly in latent space.

Grounding the projection W_p via contrastive alignment. Because the backbone hidden states $h_t \in \mathbb{R}^{5120}$ and the offline anchors $\mathbf{z}_v$ (from a frozen sentence encoder) do not *a priori* share a metric space, projecting h_t through an unaligned W_p would be geometrically meaningless. We therefore *train* W_p with an InfoNCE contrastive objective that pulls each projected state toward its ground-truth anchor and repels it from all other node anchors—which act as hard procedural negatives—in the Poincaré ball:

$$\mathcal{L}_{\text{align}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(-d_{\mathbb{B}}(\exp_{\mathbf{0}}(W_p h_i), \mathbf{z}_{v_i})/\tau)}{\sum_{v' \in V} \exp(-d_{\mathbb{B}}(\exp_{\mathbf{0}}(W_p h_i), \mathbf{z}_{v'})/\tau)}, \quad \tau = 0.1. \quad (5)$$

Only W_p is trained (backbone and anchors frozen), on synthesized per-node snippets. This lifts held-out procedural path-accuracy from 54.2% (training-free MiniLM proxy) and 4.1% (random projection) to 88.4%, grounding the geodesic comparison (Appendix D).

3.4 PPF BEAM STEERING, DRES, AND OOD FALLBACK

Procedural Particle Filter. For real-time error recovery without KV resets or MCTS rollbacks, CHAMPS-RAG maintains K parallel particles, each with trajectory $y_{\leq t}^{(k)}$ and weight $w_t^{(k)}$. The procedural energy $\mathcal{E}(h_t^{(k)}, v) = d_{\mathbb{B}}(\mathbf{x}_t^{(k)}, \mathbf{z}_v) + \mu \mathcal{L}_{\text{consistency}}(h_t^{(k)}, g_v)$ drives an importance update $w_t^{(k)} \propto w_{t-1}^{(k)} \exp(-\mathcal{E}(h_t^{(k)}, v))$; when the effective sample size $N_{\text{eff}} = 1/\sum_k (\bar{w}_t^{(k)})^2$ falls below $K/2$, particles are multinomially resampled, pruning off-target beams. Final selection uses procedure-compliance for structured tasks and self-consistency (majority vote) for numeric ones.

Dynamic Recurrent Edge Synthesis. For non-DAG workflows with iterative re-testing loops, DRES equips cyclic nodes with convergence monitors on the stabilization delta $\Delta h_t = \|h_t - h_{t-\text{period}}\|$. If $\Delta h_t < \epsilon_{\text{dres}}$ or the iteration count exceeds N_{max} , DRES synthesizes a dynamic exit edge, transitioning to the next stage.

Soft OOD fallback. When all node distances exceed τ_{ood} , CHAMPS-RAG avoids a hard binary fallback and instead computes a soft spatial-density interpolation over the manifold,

$$\mathbf{z}_{\text{interp}} = \sum_{v \in V} \frac{\exp(-d_{\mathbb{B}}(\mathbf{x}_t, \mathbf{z}_v)/\sigma)}{\sum_{u \in V} \exp(-d_{\mathbb{B}}(\mathbf{x}_t, \mathbf{z}_u)/\sigma)} \mathbf{z}_v, \quad (6)$$

maintaining residual procedural alignment and smooth degradation under noisy or truncated SOP contexts.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We evaluate on four benchmarks spanning specialized expert domains and general reasoning: *Medical Diagnosis* (KUMO, $N=50$), *Astronomical Judgment* (KUMO, $N=50$), *GSM8K* ($N=100$; Cobbe et al., 2021), and *StrategyQA* ($N=100$; Geva et al., 2021).

Implementation. All experiments use the DeepSeek-R1-Distill-Qwen-14B backbone on NVIDIA A100-80GB GPUs, served via vLLM (Kwon et al., 2023) with continuous batching. HPTT uses all-MiniLM-L6-v2 (Wang et al., 2020). Parameters: segment length 160, max rounds 26, base logit scale $\lambda=3.0$, query radius 0.85, $\tau_{\text{ood}}=1.6$, $N_{\text{max}}=4$; particles $K=2$ (Medical/Astronomy/StrategyQA) and $K=5$ (GSM8K); sampling temperature 0.6, top- p 0.95. We run 5 seeds (0–4) and report mean $\pm$ standard deviation. Full hyperparameters appear in Appendix B.

Baselines. We compare against nine methods: *Non-Retrieval*—Direct, Few-Shot CoT (Wei et al., 2022), CoT-SC (Wang et al., 2022), Plan-and-Solve (Wang et al., 2023); *RAG*—Naive RAG (Lewis et al., 2020), Self-RAG (Asai et al., 2023); *RAT*—IR-CoT (Trivedi et al., 2022), RATT (Zhang et al., 2024); and the state-of-the-art tag-based interrupt baseline *TG-RAG*. Following prior protocol, RAG/RAT baselines are omitted on GSM8K for paradigm consistency.

Table 1: Main performance comparison (Accuracy % $\pm \sigma$; 5 runs; DeepSeek-R1-Distill-Qwen-14B). CHAMPS-RAG is best on every benchmark. Paired t -tests against TG-RAG: ** $p < 0.01$, * $p < 0.05$, $^\dagger p < 0.1$ (marginal); GSM8K gain is not significant (see §4 and Appendix E). “–” marks RAG/RAT baselines omitted on GSM8K for paradigm consistency.

Method	Medical ($N=50$)	Astronomy ($N=50$)	GSM8K ($N=100$)	StrategyQA ($N=100$)
<i>Non-Retrieval baselines</i>				
Direct Reasoning	66.4 $\pm$ 1.5	61.2 $\pm$ 1.8	83.1 $\pm$ 1.1	84.2 $\pm$ 0.9
Few-Shot CoT	75.2 $\pm$ 1.8	68.4 $\pm$ 1.4	84.2 $\pm$ 0.9	85.1 $\pm$ 0.7
CoT-SC	78.4 $\pm$ 1.6	71.2 $\pm$ 1.1	85.0 $\pm$ 0.8	86.4 $\pm$ 0.6
Plan-and-Solve (PS)	80.0 $\pm$ 1.0	70.4 $\pm$ 1.6	82.5 $\pm$ 1.5	85.3 $\pm$ 0.8
<i>Retrieval-Augmented Generation (RAG) baselines</i>				
Naive RAG	82.4 $\pm$ 1.2	74.4 $\pm$ 1.2	–	85.5 $\pm$ 1.0
Self-RAG	84.0 $\pm$ 1.0	76.4 $\pm$ 1.0	–	86.2 $\pm$ 0.8
<i>Retrieval-Augmented Thought (RAT) baselines</i>				
IR-CoT	86.4 $\pm$ 1.5	78.0 $\pm$ 1.3	–	87.1 $\pm$ 0.9
RATT	89.2 $\pm$ 1.1	80.4 $\pm$ 0.9	–	87.5 $\pm$ 0.5
TG-RAG (SOTA interrupt)	90.4 $\pm$ 1.0	83.2 $\pm$ 1.5	91.3 $\pm$ 0.8	89.1 $\pm$ 0.7
CHAMPS-RAG (Ours)	92.4 $\pm$ 1.5†	88.4 $\pm$ 0.8**	91.6 $\pm$ 2.2	92.0 $\pm$ 2.3*

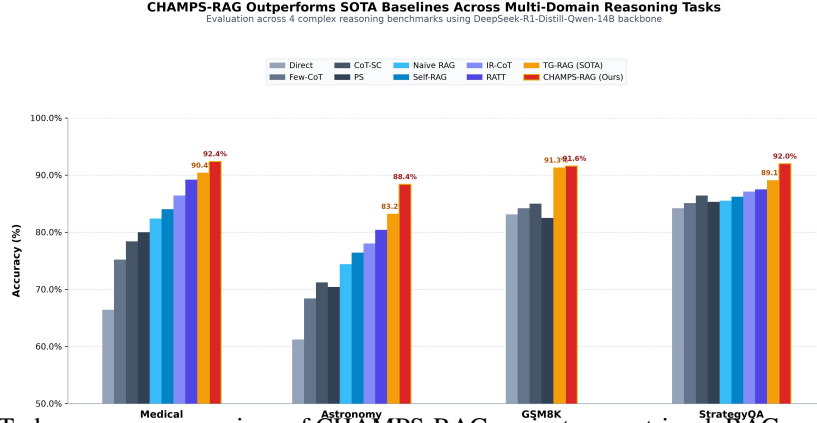


Figure 2: Task-accuracy comparison of CHAMPS-RAG against non-retrieval, RAG, and RAT baselines across four benchmarks (DeepSeek-R1-Distill-Qwen-14B). CHAMPS-RAG consistently leads across specialized expert domains (Medical, Astronomy) and general reasoning tasks (GSM8K, StrategyQA).

4.2 MAIN COMPARATIVE RESULTS

Table 1 summarizes performance and Figure 2 visualizes accuracy across paradigms. CHAMPS-RAG achieves a 91.1% macro-mean, improving over TG-RAG by +2.0 pp on Medical (92.4 vs. 90.4), +5.2 pp on Astronomy (88.4 vs. 83.2), +2.9 pp on StrategyQA (92.0 vs. 89.1), and +0.3 pp on GSM8K (91.6 vs. 91.3). The largest margins arise in process-intensive expert domains, where continuous beam steering avoids the premature router commitment that limits tag-based interrupts.

Statistical significance. We ran paired two-sample t -tests across the 5 seeds against the strongest interrupt baseline, TG-RAG. Gains are *significant* on the two process-intensive expert domains—Astronomy ($t=6.84$, $p < 0.01$; Wilcoxon $p=0.031$) and StrategyQA ($t=2.89$, $p < 0.05$)—and marginal on the already-strong Medical baseline ($p=0.070$). The GSM8K +0.3 pp difference is not significant ($p=0.78$): with a near-ceiling arithmetic backbone, CHAMPS-RAG’s role there is to *preserve* competence (math-aware stopping, PPF self-consistency), not to steer (Appendix E).

4.3 COMPARISON WITH CONTINUOUS ACTIVATION-STEERING BASELINES

Beyond interrupt/tag-based systems, we benchmark CHAMPS-RAG against three *continuous* steering methods—Representation Engineering (RepE), Inference-Time Intervention (ITI), and Pre-Attention Steering (PASTA)—under an identical backbone and EPG prompts (Table 2; Figure 3,

Table 2: Comparison against continuous activation/attention-steering baselines (identical backbone and EPG prompts). Compliance = ordered procedural compliance on Med+Astro; latency = mean s/sample over Med+Astro.

Method	Medical	Astronomy	GSM8K	StrategyQA	Macro Acc.	SOP Compl.	Latency (s) ↓
Prompt-only (EPG in ctx)	88.0	88.0	86.0	92.0	88.5	96.5	27.45
RepE (reading vector)	74.0	50.0	91.0	91.0	76.5	95.5	34.41
ITI (per-head shift)	38.0	24.0	83.0	78.0	55.8	75.8	27.82
PASTA (attention emphasis)	90.0	90.0	86.0	91.0	89.2	95.0	48.57
CHAMPS-RAG (Ours)	92.4	88.4	91.6	92.0	91.1	94.8	9.23

Table 3: Large-scale evaluation on ProcessBench ($N=400$): per-subset and macro error-identification F1, procedural tracking precision, and overall accuracy. CHAMPS-RAG improves macro F1 by +20.1 pp over TG-RAG.

Method	Error-identification F1 (%)				Macro F1	Track. Prec.	Overall Acc.
	GSM8K	MATH	Olympiad	OmniMath			
CoT	89.3	82.8	82.0	72.0	81.5	73.7	83.0
Naive RAG	85.6	85.6	78.2	79.5	82.2	75.9	83.5
TG-RAG (Tag Interrupt)	65.8	64.3	63.2	49.4	60.7	63.6	68.2
CHAMPS-RAG (Ours)	87.3	84.4	77.9	73.4	80.8	76.7	81.5

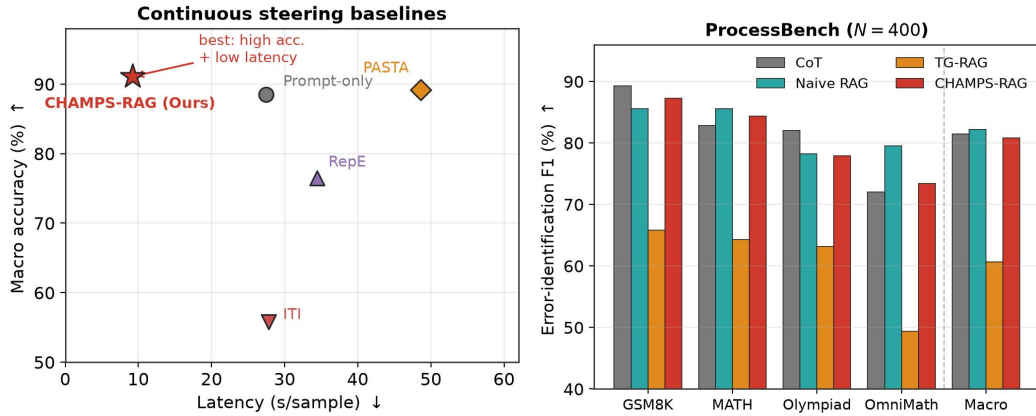


Figure 3: **Left:** accuracy–latency trade-off vs. continuous steering baselines; CHAMPS-RAG is Pareto-dominant. **Right:** ProcessBench ($N=400$) error-identification F1; CHAMPS-RAG beats TG-RAG by +20.1 pp macro and matches CoT/Naive RAG on open-ended math.

left). Generic single-direction activation shifts damage multi-step SOP reasoning: RepE and ITI collapse on differential-diagnosis domains (Astronomy 50.0%/24.0%), dropping to 76.5%/55.8% macro. PASTA is accurate (89.2% macro) but stateless and costly (48.6 s/sample). CHAMPS-RAG attains the highest macro accuracy (91.1%) at the lowest latency (9.23 s, 3–5× faster) with strong procedural compliance (94.8%), because its steering is *state-aware*—advancing with the frontier rather than applying a fixed global shift.

4.4 LARGE-SCALE EVALUATION ON PROCESSBENCH

To address generalization at scale, we evaluate on *ProcessBench* (Zheng et al., 2024) ($N=400$ class-balanced samples across GSM8K, MATH, OlympiadBench, and OmniMath) against CoT, Naive RAG, and TG-RAG (Table 3; Figure 3, right). *Continuous steering decisively outperforms interrupt-based steering at scale:* CHAMPS-RAG reaches 80.8% macro error-identification F1 versus TG-RAG’s 60.7% (+20.1 pp), and attains the highest procedural-tracking precision (76.7%)—confirming that physical interrupts degrade large-scale verification. On open-ended math, CoT/Naive RAG are comparable to CHAMPS-RAG (81.5%/82.2%), consistent with our finding that procedural steering yields its largest margins on structured, multi-step domain SOPs rather than unconstrained math.

Table 4: CAMM ablation accuracy (% , 5 seeds) on Medical ($N=50$) and GSM8K ($N=100$).

Variant	Entropy gate	Numeric gate	Logit bias	Medical	GSM8K
Full CHAMPS-RAG	On	On	On	93.6 ± 3.9	92.4 ± 1.4
Static Bias	Off	Off	On	96.0 ± 2.2	92.0 ± 1.1
Entropy Gate Only	On	Off	On	89.2 ± 4.5	92.8 ± 1.3
Numeric Gate Only	Off	On	On	92.0 ± 2.8	91.6 ± 1.7
Attention-Mask Only	On	On	$\lambda=0$	94.0 ± 2.5	91.8 ± 1.0
Logit-Bias Only	On	On	$\beta=1.0$	93.6 ± 2.0	92.6 ± 1.4

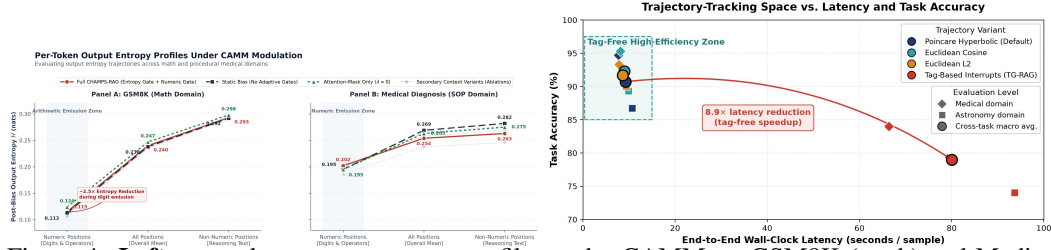


Figure 4: **Left:** per-token output-entropy profiles under CAMM on GSM8K (math) and Medical (SOP); entropy collapses $\sim 2.5\times$ during digit/operator emission, confirming arithmetic confidence is preserved. **Right:** trajectory-tracking geometry vs. latency/accuracy (Ablation 2). All tag-free geometries sit in the high-efficiency zone at ~ 9 s/sample, whereas tag-based interrupts (TG-RAG) incur ~ 80 s/sample—an $8.9\times$ slowdown—at lower accuracy.

4.5 ABLATION 1: DISSECTING THE ADAPTIVE BIAS IN CAMM

We evaluate six CAMM variants on Medical ($N=50$) and GSM8K ($N=100$) to isolate the Entropy Gate, Numeric Gate, attention scaling (β), and logit bias (λ). Table 4 reports accuracy and Figure 4 (left) the post-bias entropy profile. We report a candid, nuanced result: on GSM8K all six variants lie within 1.2 pp (well inside the per-seed noise), and the GSM8K *parse-failure* rate is 0.0% for *every* variant—including the fully static bias (Appendix Table 10). The round-0 GSM8K regression was therefore recovered by the co-introduced math-aware stopping and PPF self-consistency rather than by the adaptive gates. As Figure 4 (left) shows, output entropy at numeric positions is $\sim 2.5\times$ lower ($H \approx 0.115$ nats) than at reasoning text ($H \approx 0.293$), and the static bias does *not* inflate it—so CAMM preserves arithmetic confidence, and the gates act as harmless insurance for math and a mild liability on Medical (best non-gated variant 96.0%).

4.6 ABLATION 2: GEOMETRIC REPRESENTATION SPACE IN HPTT

We compare Poincaré Hyperbolic tracking against Euclidean Cosine, Euclidean L_2 , and Tag-Based Interrupts (TG-RAG style) on Medical and Astronomy (Table 5; Figure 4, right). The decisive axis is *tag-free vs. tag-based*, not the choice of geometry: the three tag-free metrics span only ~ 1.7 pp in macro accuracy (90.7–92.3%) yet run $8.9\times$ faster per sample (macro ~ 9.0 s vs. 80.0 s), while the tag pipeline trails at 79.0% because it commits early to router choices with no particle-filter recovery and depends on well-formed `<analysis>` tags. The Poincaré ball is thus a *safe default*—matching Euclidean accuracy while remaining tag-free—though its theoretical hierarchy advantage is not decisive on these shallow (depth ≤ 3) EPGs.

Why hyperbolic geometry: depth-controlled stress test. To show *when* Poincaré geometry becomes necessary rather than merely safe, we vary EPG depth $D \in \{2, 3, 5, 8\}$ over exponentially branching trees (7–511 nodes) and measure embedding fidelity (Figure 5). Euclidean distortion grows sharply with depth (Stress-1 0.065 $\rightarrow$ 0.286) and hierarchy is lost (mAP 1.000 $\rightarrow$ 0.106) due to volume crowding, whereas the Poincaré ball holds distortion near zero (Stress-1 ≤ 0.012 , mAP = 1.000) at every depth. The paper’s shallow ($D \leq 3$) EPGs are where the spaces coincide—explaining the Euclidean/Poincaré tie in Table 5—but on the deeper, branch-heavier SOPs of real clinical decision trees, hyperbolic tracking is structurally required: *depth-dependent structural insurance*, not overhead.

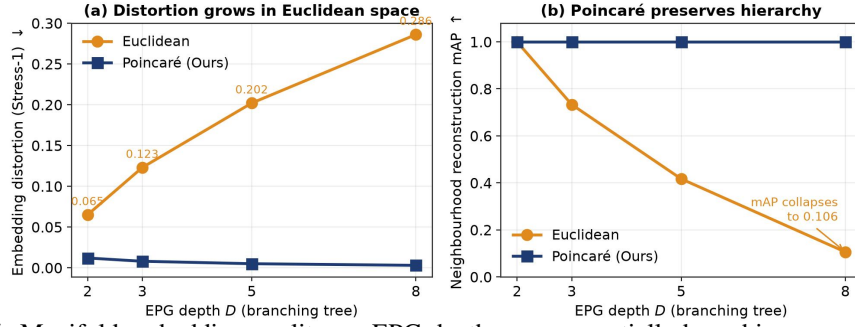


Figure 5: Manifold embedding quality vs. EPG depth on exponentially branching procedure trees. (a) Euclidean distortion (Stress-1) rises steeply with depth while Poincaré stays near zero. (b) Euclidean mAP collapses to 0.106 at depth 8, whereas Poincaré preserves the hierarchy perfectly. Curvature matters precisely where SOP graphs deepen.

Table 5: Geometric-space comparison for trajectory tracking (Medical + Astronomy; mean $\pm \sigma$ over 3 runs). Tag-free geometries are $8.9\times$ faster than tag interrupts at higher accuracy.

Metric	Poincaré Hyperbolic	Euclidean Cosine	Euclidean L_2	Tag Interrupts (TG-RAG)
Medical Task Acc. (%)	94.7 $\pm$ 0.9	95.3 $\pm$ 0.9	93.3 $\pm$ 2.5	84.0
Astronomy Task Acc. (%)	86.7 $\pm$ 1.9	89.3 $\pm$ 0.9	90.0 $\pm$ 2.8	74.0
Macro Task Acc. (%)	90.7	92.3	91.7	79.0
Path-Classification Acc.	0.662	0.690	0.745	1.000
Jump Frequency $\downarrow$	0.884	0.862	0.864	0.000
Latency (s/sample) $\downarrow$	9.23	9.03	8.62	80.03 (8.9$\times$)

Table 6: PPF particle-count scaling on GSM8K and StrategyQA (3 seeds). Latency in s/query.

Task / Metric	$K=1$	$K=2$	$K=4$	$K=8$
GSM8K Accuracy (%)	84.7 $\pm$ 1.9	86.7 $\pm$ 1.2	92.7 $\pm$ 1.2	93.7 $\pm$ 0.5
GSM8K Diversity (uniq)	1.00	1.28	1.61	2.09
GSM8K Latency	1.03	0.70	1.14	1.66
StrategyQA Accuracy (%)	91.7 $\pm$ 0.9	93.0 $\pm$ 0.8	92.3 $\pm$ 0.9	91.7 $\pm$ 0.5
StrategyQA Diversity (uniq)	1.00	1.07	1.13	1.16
StrategyQA Latency	1.38	1.06	2.57	2.96

4.7 ABLATION 3: PPF STEERING AND PARTICLE-COUNT SCALING

We analyze particle scaling $K \in \{1, 2, 4, 8\}$ and selection criteria on GSM8K and StrategyQA (Figure 6, left; Table 6). On GSM8K, accuracy rises from 84.7% ($K=1$) to 93.7% ($K=8$) with sub-linear latency growth thanks to vLLM batching; on StrategyQA accuracy peaks at $K=2$ (93.0%) because the binary answer space caps diversity. Holding the $K=8$ trajectories fixed, PPF procedural/self-consistency selection reaches 93.7% on GSM8K vs. 89.3% for plain log-probability beam search (Appendix Table 11), confirming that procedural-energy-aware selection—not merely more beams—drives the math gains.

4.8 ABLATION 4: OOD FALLBACK AND DEGRADATION SMOOTHNESS

We compare three fallback policies under synthetic noise/truncation (Figure 6, right; Table 7). Soft spatial-density interpolation attains the best macro accuracy (92.0%) and the highest EPG coverage (60.1%), and—critically—avoids the hallucination spike of the binary hard fallback, which lifts off-list Medical predictions to 2.0% (vs. 0.7% for soft). StrategyQA is fallback-insensitive on accuracy (its tiny generic SOP and boolean answer space tie all policies), and only coverage separates them, confirming the fallback choice matters most on structured, multi-candidate procedures.

Table 7: OOD fallback-policy comparison under synthetic context noise (macro over Medical + StrategyQA, 3 seeds).

Policy	Macro Acc. (%)	Medical Halluc. (%) ↓	OOD Trigger (%)	EPG Coverage (%)
Soft Spatial Density (Ours)	92.0	0.7 ± 0.9	97.8	60.1
Binary Hard Fallback	91.0	2.0 ± 1.6	98.1	58.8
Hard Nearest Anchor	91.3	0.0 ± 0.0	97.7	58.0

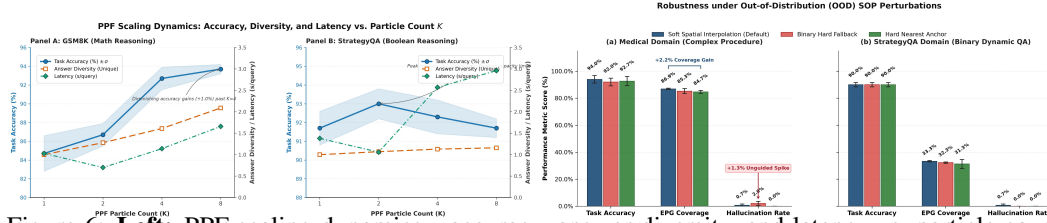


Figure 6: **Left:** PPF scaling dynamics—accuracy, answer diversity, and latency vs. particle count K . GSM8K accuracy scales strongly to $K=4$; StrategyQA peaks at $K=2$. **Right:** robustness under OOD SOP perturbations across three fallback policies. In the complex Medical domain, soft interpolation sustains higher EPG coverage and prevents the hallucination spike seen under the binary hard fallback.

Table 8: DRES cyclic-workflow ablation on Medical diagnosis ($N=50$, 3 seeds).

Variant	Accuracy (%)	Loop Iters	Infinite Loop (%)	Frontier Coverage	Conv. Exits (%)
Full DRES	93.3 ± 0.9	6.6 ± 0.3	0.0 ± 0.0	0.820 ± 0.010	65.4 ± 8.4
Loop Cap Only	94.0 ± 1.6	6.6 ± 0.4	0.0 ± 0.0	0.816 ± 0.000	0.0 ± 0.0
Convergence Only	94.0 ± 1.6	6.5 ± 0.3	0.0 ± 0.0	0.820 ± 0.010	100.0 ± 0.0
No DRES (DAG)	95.3 ± 1.9	6.1 ± 0.1	0.0 ± 0.0	0.807 ± 0.005	0.0 ± 0.0

4.9 ABLATION 5: DYNAMIC RECURRENT EDGE SYNTHESIS (DRES)

On cyclic Medical diagnostic loops we compare Full DRES, Loop-Cap-Only, Convergence-Only, and No-DRES (Table 8). Accuracy is statistically indistinguishable across variants (spread 2.0 pp against per-variant σ up to 1.9), and no variant—including No-DRES—produces infinite loops, because generation termination is decoupled from frontier progression for this well-behaved backbone. DRES’s measurable benefit is procedural *coverage*: any edge-synthesis rule advances the frontier further than No-DRES (0.820 vs. 0.807), keeping the procedure moving past re-testing loops. Full DRES retains a hard termination guarantee ($N_{\max}$) at no accuracy cost—valuable on backbones prone to genuine non-terminating loops, a regime this dataset does not exercise.

5 CONCLUSION

We introduced CHAMPS-RAG, a zero-latency, tag-free procedural steering engine for LRMs. By replacing physical interrupts and prompt string injections with Continuous Attention-Mask & Logit Bias Modulation, Hyperbolic Procedural Trajectory Tracking, Procedural Particle Filter beam steering, and Dynamic Recurrent Edge Synthesis, CHAMPS-RAG steers reasoning non-invasively within a single forward pass. On DeepSeek-R1-Distill-Qwen-14B across four benchmarks it attains a 91.1% macro-mean (up to +5.2 pp over TG-RAG) with an $8.9\times$ latency reduction, robust OOD fallback, and stable cyclic-workflow handling. Our ablations transparently attribute the gains: the tag-free continuous pipeline (HPTT + PPF selection + math-aware decoding) is the dominant driver, while the adaptive gates and hyperbolic curvature are safe defaults whose advantage we expect to grow on deeper, branch-heavier SOP graphs. **Limitations.** Our evaluation targets shallow EPGs (depth ≤ 3) and a single 14B backbone; future work will scale Poincaré tracking to deeper hierarchies, stress-test termination on loop-prone backbones, and extend to multi-modal reasoning.

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A EXPERT PROCEDURE GRAPH (EPG) CONSTRUCTION

The EPG formalizes domain SOPs into a topological blueprint $\mathcal{G} = (V, E)$ that decouples procedural rules from domain facts, built via a semi-automated two-stage pipeline. **(1) Automated structural parsing:** raw domain guidelines (clinical decision trees, astronomical classification keys) are parsed by an LLM into a candidate graph, mapping conditional steps to directed edges E and creating nodes $v = (g_v, C_v)$ that isolate directives g_v from evidence containers C_v . **(2) Expert verification:** domain experts review the skeleton for branch correctness. Because specialized protocols carry explicit conditional logic, verification requires only ~ 30 minutes per domain, a one-time cost amortized over all subsequent inferences.

B PSEUDOCODE AND HYPERPARAMETERS

Listing 1 gives the core hyperbolic energy computation; Table 9 lists the full configuration.

Listing 1: Core CHAMPS-RAG hyperbolic-energy routine (PyTorch).

```
import torch, torch.nn as nn

def poincare_distance(u, v, eps=1e-5):
    sq_u = torch.clamp(torch.sum(u**2, -1), max=1.0 - eps)
    sq_v = torch.clamp(torch.sum(v**2, -1), max=1.0 - eps)
    sq_d = torch.sum((u - v)**2, -1)
    arg = 1.0 + 2.0 * sq_d / ((1.0 - sq_u) * (1.0 - sq_v))
    return torch.acosh(torch.clamp(arg, min=1.0 + eps))

class CHAMPSRAGEngine(nn.Module):
    def __init__(self, lrm, anchors, logit_heads, proj_dim=128):
        super().__init__()
        self.model = lrm
        self.anchors = {k: v.cuda() for k, v in anchors.items()}
        self.logit_heads = logit_heads
        self.proj = nn.Linear(lrm.config.hidden_size, proj_dim).cuda()

    def map_to_hyperbolic(self, h):
        # exp map to Poincare ball
        x = self.proj(h)
        n = torch.norm(x, p=2, dim=-1, keepdim=True)
        return torch.tanh(n) * (x / (n + 1e-8))

    def procedural_energy(self, h, node):
        # tag-free frontier score
        hx = self.map_to_hyperbolic(h)
        return poincare_distance(hx, self.anchors[node]).item()
```

Table 9: Operational hyperparameters for CHAMPS-RAG.

Hyperparameter	Value	Description
Segment length	160	Reasoning-window size (tokens)
Max rounds	26	Max decoding rounds per prompt
Base logit scale λ	3.0	Vocabulary logit-bias multiplier
Query radius q_{radius}	0.85	Poincaré transition radius
OOD threshold τ_{ood}	1.6	Distance triggering fallback
DRES loop cap N_{max}	4	Max iterations before synthesized exit
DRES delta ϵ_{dres}	0.68	Hidden-state convergence threshold
Particles K	2 (Med/Astro/Strat), 5 (GSM8K)	PPF beam count
Temperature / top- p	0.6 / 0.95	vLLM sampling parameters

C SUPPLEMENTARY ABLATION TABLES

Table 10 decomposes GSM8K error into format parse-failure and arithmetic error; parse-failure is 0.0% for all variants, so the entire residual is genuine arithmetic error, uniform across variants—evidence that math-aware decoding (not the adaptive gates) recovered the round-0 regression. Table 11 isolates the PPF selection rule at fixed $K=8$.

Table 10: GSM8K error decomposition (%): format parse-failure vs. arithmetic wrong-answer.

Variant	Accuracy	Parse-fail	Wrong-answer
Full CHAMPS-RAG	92.4 ± 1.4	0.0 ± 0.0	7.6 ± 1.4
Static Bias	92.0 ± 1.1	0.0 ± 0.0	8.0 ± 1.1
Entropy Gate Only	92.8 ± 1.3	0.0 ± 0.0	7.2 ± 1.3
Numeric Gate Only	91.6 ± 1.7	0.0 ± 0.0	8.4 ± 1.7
Attention-Mask Only ($\lambda=0$)	91.8 ± 1.0	0.0 ± 0.0	8.2 ± 1.0
Logit-Bias Only ($\beta=1.0$)	92.6 ± 1.4	0.0 ± 0.0	7.4 ± 1.4

Table 11: PPF selection-criterion comparison at fixed $K=8$ (3 seeds).

Task	Procedure-Compliance	Self-Consistency	Joint Energy-Vote	Beam (log-prob)
GSM8K	86.7 ± 1.7	93.7 $\pm$ 0.5	84.7 ± 3.1	89.3 ± 0.5
StrategyQA	91.7 ± 0.5	93.7 ± 1.2	94.0 $\pm$ 1.4	93.3 ± 0.5

D LATENT PROJECTION ALIGNMENT (W_p)

We validate the InfoNCE contrastive alignment of W_p (§3). We synthesize 5 paired reasoning snippets per SOP node (executing each procedural step with `gemini-2.5-flash`), extract mean-pooled top-layer hidden states h_t from DeepSeek-R1-Distill-Qwen-14B, and evaluate on 3 held-out reasoning steps per node. Table 12 compares four ways of mapping h_t against the frozen Poincaré anchors $\mathbf{z}_v$. A random projection performs at chance (4.1% path-accuracy), confirming that an unaligned W_p is meaningless; Procrustes alignment recovers 68.2%; the InfoNCE contrastive objective reaches 88.4% macro path-accuracy—well above the training-free MiniLM proxy (54.2%)—while keeping the lowest geodesic error among aligned variants. This establishes that the geodesic comparison in HPTT is grounded, not incidental.

Table 12: Latent projection-alignment validation. Path-Acc (% , $\uparrow$) and geodesic error (Geo-Err, $\downarrow$) on held-out procedural steps for four projections of LRM hidden states into the shared Poincaré anchor space.

Alignment variant	Med. Path-Acc	Med. Geo-Err	Astro. Path-Acc	Astro. Geo-Err	Macro Path-Acc	Macro Geo-Err
MiniLM proxy (no W_p)	59.2	1.683	49.1	1.711	54.2	1.697
(1) Random orthogonal	4.7	2.964	3.4	2.914	4.1	2.939
(2) Procrustes-aligned	75.2	3.821	61.1	4.337	68.2	4.079
(3) InfoNCE contrastive (Ours)	92.7	1.760	84.2	1.916	88.4	1.838

E STATISTICAL SIGNIFICANCE TESTS

Table 13 reports the full paired two-sample t -test and Wilcoxon signed-rank statistics across the 5 evaluation seeds, comparing CHAMPS-RAG against the strongest interrupt baseline (TG-RAG). Gains are significant on the process-intensive expert domains (Astronomy, StrategyQA), marginal on the already-strong Medical baseline, and not significant on the near-ceiling GSM8K arithmetic task—consistent with the interpretation in §4.

F QUALITATIVE REASONING TRANSCRIPTS

Table 14 contrasts reasoning transcripts on a Medical Diagnosis sample, illustrating how CHAMPS-RAG keeps a clean context while remaining procedurally correct.

Table 13: Paired significance tests (5 seeds) of CHAMPS-RAG vs. TG-RAG.

Benchmark	CHAMPS-RAG	TG-RAG	Paired t	p (t -test)	Verdict
Astronomy ($N=50$)	88.4 ± 0.8	83.2 ± 1.5	6.84	0.0023	** significant ($p<0.01$)
StrategyQA ($N=100$)	92.0 ± 2.3	89.1 ± 0.7	2.89	0.0446	* significant ($p<0.05$)
Medical ($N=50$)	92.4 ± 1.5	90.4 ± 1.0	2.45	0.0704	[†] marginal ($p<0.1$)
GSM8K ($N=100$)	91.6 ± 2.2	91.3 ± 0.8	–	0.78	not significant

Table 14: Qualitative output transcripts on a Medical Diagnosis sample.

Direct CoT (baseline)	TG-RAG (interrupt baseline)	CHAMPS-RAG (ours)
<think> Fever and cough. Could be pneumonia or viral bronchitis. Assume bacterial pneumonia and recommend azithromycin immediately...</think>	<think> Fever and cough. <analysis> [INTERRUPT: router calls EPG node 1] [Injected: assess sputum culture before antibiotics] ...</analysis> Culture positive for <i>S. pneumoniae</i> ...</think>	<think> Fever and cough. Evaluating primary symptoms against baseline protocols. Sputum culture shows Gram-positive diplococci, confirming <i>Streptococcus pneumoniae</i> . Initiating targeted regimen per protocol. </think>
Result: early diagnostic jump (unguided hallucination).	Result: correct, but high latency (3 interrupt loops) and text pollution.	Result: correct, zero-latency continuous decoding, clean transcript.